

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

## Topic 30 — Hydrocarbons: Arenes (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| CANDIDATE      | Urwah                                                                    |
| SYLLABUS       | Cambridge International A Level Chemistry (9701)                         |
| TOPIC COVERAGE | 110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis |
| QUESTION COUNT | 110 Questions (100 Comprehensive + 10 High-Frequency Repeats)            |
| TOTAL MARKS    | 110 Marks (1 Mark Each · Standardized A Level Scoring)                   |

### HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

### SYLLABUS SUBTOPICS COVERED

|              |                 |
|--------------|-----------------|
| Core MCQs    | — 100 Questions |
| Core Repeats | — 10 Questions  |

## MCQs

**1 Enthalpy of Hydrogenation of Benzene — 9701/11/M/J/23/Q21**

HARD · Syllabus 30.1

The enthalpy of hydrogenation of cyclohexene is  $-120 \text{ kJ mol}^{-1}$ . The experimental enthalpy of hydrogenation of benzene is  $-208 \text{ kJ mol}^{-1}$ . What is the resonance stabilization energy of benzene?

- A:  $360 \text{ kJ mol}^{-1}$
- B:  $152 \text{ kJ mol}^{-1}$
- C:  $88 \text{ kJ mol}^{-1}$
- D:  $208 \text{ kJ mol}^{-1}$

**2 Nitronium Ion Generation — 9701/12/M/J/23/Q21**

EASY · Syllabus 30.1

What is the active electrophile in the nitration of benzene by a mixture of concentrated  $\text{HNO}_3$  and concentrated  $\text{H}_2\text{SO}_4$ ?

- A:  $\text{NO}_3^-$
- B:  $\text{NO}^+$
- C:  $\text{HNO}_2$
- D:  $\text{NO}_2^+$

**3 Arenes Reaction MCQ 3 — 9701/1/22/Q3**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )
- B:  $\text{FeCl}_2$
- C:  $\text{H}_2\text{SO}_4$
- D: Ni

**4 Arenes Reaction MCQ 4 — 9701/1/22/Q4**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{H}_2\text{SO}_4$
- C:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )
- D: Ni

**5 Arenes Reaction MCQ 5 — 9701/1/22/Q5**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )
- B:  $\text{FeCl}_2$
- C:  $\text{H}_2\text{SO}_4$
- D: Ni

**6 Arenes Reaction MCQ 6 — 9701/1/22/Q6**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{H}_2\text{SO}_4$
- C: Ni
- D:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )

**7 Arenes Reaction MCQ 7 — 9701/1/22/Q7**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**8 Arenes Reaction MCQ 8 — 9701/1/22/Q8**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**9 Arenes Reaction MCQ 9 — 9701/1/22/Q9**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**10 Arenes Reaction MCQ 10 — 9701/1/22/Q10**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**11 Arenes Reaction MCQ 11 — 9701/1/22/Q11**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**12 Arenes Reaction MCQ 12 — 9701/1/22/Q12**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**13 Arenes Reaction MCQ 13 — 9701/1/22/Q13**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**14 Arenes Reaction MCQ 14 — 9701/1/22/Q14**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**15 Arenes Reaction MCQ 15 — 9701/1/22/Q15**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**16 Arenes Reaction MCQ 16 — 9701/1/22/Q16**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**17 Arenes Reaction MCQ 17 — 9701/1/22/Q17**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**18 Arenes Reaction MCQ 18 — 9701/1/22/Q18**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**19 Arenes Reaction MCQ 19 — 9701/1/22/Q19**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**20 Arenes Reaction MCQ 20 — 9701/1/22/Q20**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**21 Arenes Reaction MCQ 21 — 9701/1/22/Q21**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**22 Arenes Reaction MCQ 22 — 9701/1/22/Q22**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**23 Arenes Reaction MCQ 23 — 9701/1/22/Q23**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**24 Arenes Reaction MCQ 24 — 9701/1/22/Q24**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**25 Arenes Reaction MCQ 25 — 9701/1/22/Q25**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**26 Arenes Reaction MCQ 26 — 9701/1/22/Q26**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**27 Arenes Reaction MCQ 27 — 9701/1/22/Q27**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**28 Arenes Reaction MCQ 28 — 9701/1/22/Q28**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**29 Arenes Reaction MCQ 29 — 9701/1/22/Q29**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**30 Arenes Reaction MCQ 30 — 9701/1/22/Q30**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**31 Arenes Reaction MCQ 31 — 9701/1/22/Q31**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**32 Arenes Reaction MCQ 32 — 9701/1/22/Q32**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**33 Arenes Reaction MCQ 33 — 9701/1/22/Q33**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**34 Arenes Reaction MCQ 34 — 9701/1/22/Q34**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**35 Arenes Reaction MCQ 35 — 9701/1/22/Q35**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**36 Arenes Reaction MCQ 36 — 9701/1/22/Q36**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**37 Arenes Reaction MCQ 37 — 9701/1/22/Q37**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**38 Arenes Reaction MCQ 38 — 9701/1/22/Q38**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**39 Arenes Reaction MCQ 39 — 9701/1/22/Q39**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**40 Arenes Reaction MCQ 40 — 9701/1/22/Q40**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**41 Arenes Reaction MCQ 41 — 9701/1/22/Q41**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**42 Arenes Reaction MCQ 42 — 9701/1/22/Q42**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**43 Arenes Reaction MCQ 43 — 9701/1/22/Q43**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**44 Arenes Reaction MCQ 44 — 9701/1/22/Q44**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**45 Arenes Reaction MCQ 45 — 9701/1/22/Q45**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**46 Arenes Reaction MCQ 46 — 9701/1/22/Q46**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**47 Arenes Reaction MCQ 47 — 9701/1/22/Q47**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**48 Arenes Reaction MCQ 48 — 9701/1/22/Q48**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**49 Arenes Reaction MCQ 49 — 9701/1/22/Q49**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**50 Arenes Reaction MCQ 50 — 9701/1/22/Q50**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**51 Arenes Reaction MCQ 51 — 9701/1/22/Q51**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**52 Arenes Reaction MCQ 52 — 9701/1/22/Q52**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**53 Arenes Reaction MCQ 53 — 9701/1/22/Q53**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**54 Arenes Reaction MCQ 54 — 9701/1/22/Q54**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**55 Arenes Reaction MCQ 55 — 9701/1/22/Q55**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**56 Arenes Reaction MCQ 56 — 9701/1/22/Q56****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{H}_2\text{SO}_4$
- C: Ni
- D:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )

**57 Arenes Reaction MCQ 57 — 9701/1/22/Q57****EASY · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{H}_2\text{SO}_4$
- C:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )
- D: Ni

**58 Arenes Reaction MCQ 58 — 9701/1/22/Q58****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )
- B:  $\text{FeCl}_2$
- C:  $\text{H}_2\text{SO}_4$
- D: Ni

**59 Arenes Reaction MCQ 59 — 9701/1/22/Q59****EASY · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{H}_2\text{SO}_4$
- C: Ni
- D:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )

**60 Arenes Reaction MCQ 60 — 9701/1/22/Q60****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )
- C:  $\text{H}_2\text{SO}_4$
- D: Ni

**61 Arenes Reaction MCQ 61 — 9701/1/22/Q61****EASY · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )
- C:  $\text{H}_2\text{SO}_4$
- D: Ni

**62 Arenes Reaction MCQ 62 — 9701/1/22/Q62****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A:  $\text{FeCl}_2$
- B:  $\text{H}_2\text{SO}_4$
- C: Ni
- D:  $\text{FeBr}_3$  (or  $\text{AlBr}_3$ )

**63 Arenes Reaction MCQ 63 — 9701/1/22/Q63**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**64 Arenes Reaction MCQ 64 — 9701/1/22/Q64**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**65 Arenes Reaction MCQ 65 — 9701/1/22/Q65**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**66 Arenes Reaction MCQ 66 — 9701/1/22/Q66**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**67 Arenes Reaction MCQ 67 — 9701/1/22/Q67**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**68 Arenes Reaction MCQ 68 — 9701/1/22/Q68**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**69 Arenes Reaction MCQ 69 — 9701/1/22/Q69**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**70 Arenes Reaction MCQ 70 — 9701/1/22/Q70****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**71 Arenes Reaction MCQ 71 — 9701/1/22/Q71****EASY · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**72 Arenes Reaction MCQ 72 — 9701/1/22/Q72****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**73 Arenes Reaction MCQ 73 — 9701/1/22/Q73****EASY · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**74 Arenes Reaction MCQ 74 — 9701/1/22/Q74****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**75 Arenes Reaction MCQ 75 — 9701/1/22/Q75****EASY · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**76 Arenes Reaction MCQ 76 — 9701/1/22/Q76****HARD · Syllabus 30.1**

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**77 Arenes Reaction MCQ 77 — 9701/1/22/Q77**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**78 Arenes Reaction MCQ 78 — 9701/1/22/Q78**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**79 Arenes Reaction MCQ 79 — 9701/1/22/Q79**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**80 Arenes Reaction MCQ 80 — 9701/1/22/Q80**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**81 Arenes Reaction MCQ 81 — 9701/1/22/Q81**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**82 Arenes Reaction MCQ 82 — 9701/1/22/Q82**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**83 Arenes Reaction MCQ 83 — 9701/1/22/Q83**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**84 Arenes Reaction MCQ 84 — 9701/1/22/Q84**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**85 Arenes Reaction MCQ 85 — 9701/1/22/Q85**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**86 Arenes Reaction MCQ 86 — 9701/1/22/Q86**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**87 Arenes Reaction MCQ 87 — 9701/1/22/Q87**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**88 Arenes Reaction MCQ 88 — 9701/1/22/Q88**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**89 Arenes Reaction MCQ 89 — 9701/1/22/Q89**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**90 Arenes Reaction MCQ 90 — 9701/1/22/Q90**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**91 Arenes Reaction MCQ 91 — 9701/1/22/Q91**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**92 Arenes Reaction MCQ 92 — 9701/1/22/Q92**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**93 Arenes Reaction MCQ 93 — 9701/1/22/Q93**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**94 Arenes Reaction MCQ 94 — 9701/1/22/Q94**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**95 Arenes Reaction MCQ 95 — 9701/1/22/Q95**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**96 Arenes Reaction MCQ 96 — 9701/1/22/Q96**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**97 Arenes Reaction MCQ 97 — 9701/1/22/Q97**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- D: Ni

**98 Arenes Reaction MCQ 98 — 9701/1/22/Q98**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- B: FeCl<sub>2</sub>
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**99 Arenes Reaction MCQ 99 — 9701/1/22/Q99**

EASY · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: H<sub>2</sub>SO<sub>4</sub>
- C: Ni
- D: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)

**100 Arenes Reaction MCQ 100 — 9701/1/22/Q100**

HARD · Syllabus 30.1

Which reagent is used as a halogen carrier catalyst for the bromination of benzene?

- A: FeCl<sub>2</sub>
- B: FeBr<sub>3</sub> (or AlBr<sub>3</sub>)
- C: H<sub>2</sub>SO<sub>4</sub>
- D: Ni

**101 Core Repeat: Arenes Chemistry 101 — 9701/1/23/Q101**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline KMnO<sub>4</sub> and subsequently acidified?

- A: Phenylethanoic acid
- B: 2-phenylethanol
- C: Benzoic acid
- D: 1-phenylethanol

**102 Core Repeat: Arenes Chemistry 102 — 9701/1/23/Q102**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline KMnO<sub>4</sub> and subsequently acidified?

- A: Benzoic acid
- B: Phenylethanoic acid
- C: 2-phenylethanol
- D: 1-phenylethanol

**103 Core Repeat: Arenes Chemistry 103 — 9701/1/23/Q103**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline KMnO<sub>4</sub> and subsequently acidified?

- A: Phenylethanoic acid
- B: 2-phenylethanol
- C: 1-phenylethanol
- D: Benzoic acid

**104 Core Repeat: Arenes Chemistry 104 — 9701/1/23/Q104**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline KMnO<sub>4</sub> and subsequently acidified?

- A: Phenylethanoic acid
- B: Benzoic acid
- C: 2-phenylethanol
- D: 1-phenylethanol

**105 Core Repeat: Arenes Chemistry 105 — 9701/1/23/Q105**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline  $\text{KMnO}_4$  and subsequently acidified?

- A: Benzoic acid
- B: Phenylethanoic acid
- C: 2-phenylethanol
- D: 1-phenylethanol

**106 Core Repeat: Arenes Chemistry 106 — 9701/1/23/Q106**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline  $\text{KMnO}_4$  and subsequently acidified?

- A: Phenylethanoic acid
- B: 2-phenylethanol
- C: Benzoic acid
- D: 1-phenylethanol

**107 Core Repeat: Arenes Chemistry 107 — 9701/1/23/Q107**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline  $\text{KMnO}_4$  and subsequently acidified?

- A: Phenylethanoic acid
- B: Benzoic acid
- C: 2-phenylethanol
- D: 1-phenylethanol

**108 Core Repeat: Arenes Chemistry 108 — 9701/1/23/Q108**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline  $\text{KMnO}_4$  and subsequently acidified?

- A: Phenylethanoic acid
- B: 2-phenylethanol
- C: 1-phenylethanol
- D: Benzoic acid

**109 Core Repeat: Arenes Chemistry 109 — 9701/1/23/Q109**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline  $\text{KMnO}_4$  and subsequently acidified?

- A: Benzoic acid
- B: Phenylethanoic acid
- C: 2-phenylethanol
- D: 1-phenylethanol

**110 Core Repeat: Arenes Chemistry 110 — 9701/1/23/Q110**

HARD · Syllabus 30.1

Which compound is formed when ethylbenzene is refluxed with alkaline  $\text{KMnO}_4$  and subsequently acidified?

- A: Phenylethanoic acid
- B: 2-phenylethanol
- C: Benzoic acid
- D: 1-phenylethanol

**Quick-Check Answer Key Grid — Topic 30 — Hydrocarbons: Arenes (A Level MCQs)**

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

| Q#  | ANS | Q#  | ANS | Q#  | ANS | Q#  | ANS | Q#   | ANS |
|-----|-----|-----|-----|-----|-----|-----|-----|------|-----|
| Q1  | B   | Q23 | A   | Q45 | A   | Q67 | B   | Q89  | B   |
| Q2  | D   | Q24 | C   | Q46 | D   | Q68 | C   | Q90  | A   |
| Q3  | A   | Q25 | A   | Q47 | B   | Q69 | B   | Q91  | D   |
| Q4  | C   | Q26 | D   | Q48 | C   | Q70 | A   | Q92  | C   |
| Q5  | A   | Q27 | B   | Q49 | B   | Q71 | D   | Q93  | A   |
| Q6  | D   | Q28 | C   | Q50 | A   | Q72 | C   | Q94  | C   |
| Q7  | B   | Q29 | B   | Q51 | D   | Q73 | A   | Q95  | B   |
| Q8  | C   | Q30 | A   | Q52 | C   | Q74 | C   | Q96  | D   |
| Q9  | B   | Q31 | D   | Q53 | A   | Q75 | B   | Q97  | C   |
| Q10 | A   | Q32 | C   | Q54 | C   | Q76 | D   | Q98  | A   |
| Q11 | D   | Q33 | A   | Q55 | B   | Q77 | C   | Q99  | D   |
| Q12 | C   | Q34 | C   | Q56 | D   | Q78 | A   | Q100 | B   |
| Q13 | A   | Q35 | B   | Q57 | C   | Q79 | D   | Q101 | C   |
| Q14 | C   | Q36 | D   | Q58 | A   | Q80 | B   | Q102 | A   |
| Q15 | B   | Q37 | C   | Q59 | D   | Q81 | B   | Q103 | D   |
| Q16 | D   | Q38 | A   | Q60 | B   | Q82 | D   | Q104 | B   |
| Q17 | C   | Q39 | D   | Q61 | B   | Q83 | A   | Q105 | A   |
| Q18 | A   | Q40 | B   | Q62 | D   | Q84 | C   | Q106 | C   |
| Q19 | D   | Q41 | B   | Q63 | A   | Q85 | A   | Q107 | B   |
| Q20 | B   | Q42 | D   | Q64 | C   | Q86 | D   | Q108 | D   |
| Q21 | B   | Q43 | A   | Q65 | A   | Q87 | B   | Q109 | A   |
| Q22 | D   | Q44 | C   | Q66 | D   | Q88 | C   | Q110 | C   |

**Explanations & Distractor Analysis — Topic 30 — Hydrocarbons: Arenes (A Level MCQs)**

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

|    |          |                                                                                                                                                                                                                                                                         |     |
|----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | <b>B</b> | Option B is correct. Theoretical Kekulé cyclohexa-1,3,5-triene would be $3 \times (-120) = -360 \text{ kJ mol}^{-1}$ . The experimental value of $-208 \text{ kJ mol}^{-1}$ is $152 \text{ kJ mol}^{-1}$ more stable, representing the resonance delocalisation energy. | [1] |
| 2  | <b>D</b> | Option D is correct. $\text{H}_2\text{SO}_4$ protonates $\text{HNO}_3$ to generate the linear nitronium cation, $\text{NO}_2^+$ .                                                                                                                                       | [1] |
| 3  | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 4  | <b>C</b> | Option C is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 5  | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 6  | <b>D</b> | Option D is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 7  | <b>B</b> | Option B is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 8  | <b>C</b> | Option C is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 9  | <b>B</b> | Option B is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 10 | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 11 | <b>D</b> | Option D is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 12 | <b>C</b> | Option C is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 13 | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 14 | <b>C</b> | Option C is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 15 | <b>B</b> | Option B is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 16 | <b>D</b> | Option D is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 17 | <b>C</b> | Option C is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 18 | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 19 | <b>D</b> | Option D is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 20 | <b>B</b> | Option B is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 21 | <b>B</b> | Option B is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 22 | <b>D</b> | Option D is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 23 | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 24 | <b>C</b> | Option C is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 25 | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 26 | <b>D</b> | Option D is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 27 | <b>B</b> | Option B is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 28 | <b>C</b> | Option C is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 29 | <b>B</b> | Option B is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 30 | <b>A</b> | Option A is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |
| 31 | <b>D</b> | Option D is correct. $\text{FeBr}_3$ polarises $\text{Br}-\text{Br}$ by accepting a bromide ion to generate the $\text{Br}^+$ electrophile.                                                                                                                             | [1] |

|    |          |                                                                                                                                 |     |
|----|----------|---------------------------------------------------------------------------------------------------------------------------------|-----|
| 32 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 33 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 34 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 35 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 36 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 37 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 38 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 39 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 40 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 41 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 42 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 43 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 44 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 45 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 46 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 47 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 48 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 49 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 50 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 51 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 52 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 53 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 54 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 55 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 56 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 57 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 58 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 59 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 60 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 61 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 62 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 63 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 64 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |

|    |          |                                                                                                                                 |     |
|----|----------|---------------------------------------------------------------------------------------------------------------------------------|-----|
| 65 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 66 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 67 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 68 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 69 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 70 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 71 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 72 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 73 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 74 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 75 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 76 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 77 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 78 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 79 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 80 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 81 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 82 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 83 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 84 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 85 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 86 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 87 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 88 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 89 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 90 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 91 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 92 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 93 | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 94 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 95 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 96 | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |
| 97 | <b>C</b> | Option C is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile. | [1] |

|     |          |                                                                                                                                                                                |     |
|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 98  | <b>A</b> | Option A is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile.                                                | [1] |
| 99  | <b>D</b> | Option D is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile.                                                | [1] |
| 100 | <b>B</b> | Option B is correct. FeBr <sub>3</sub> polarises Br-Br by accepting a bromide ion to generate the Br <sup>+</sup> electrophile.                                                | [1] |
| 101 | <b>C</b> | Option C is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 102 | <b>A</b> | Option A is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 103 | <b>D</b> | Option D is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 104 | <b>B</b> | Option B is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 105 | <b>A</b> | Option A is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 106 | <b>C</b> | Option C is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 107 | <b>B</b> | Option B is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 108 | <b>D</b> | Option D is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 109 | <b>A</b> | Option A is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |
| 110 | <b>C</b> | Option C is correct. Side-chain oxidation of any alkylbenzene with at least one benzylic hydrogen cleaves the alkyl chain down to a single -COOH group, yielding benzoic acid. | [1] |